Combinations KEY

Name _____

Pick 3 students from 5 (A, B, C, D, E) to be on a committee. All 3 have the same job. Choosing A, B, C is the **same committee** as choosing C, A, B. **Order does not matter.**

Count as if order mattered: ${}_5P_3 = 5 \cdot 4 \cdot 3 = 60$
Each committee of 3 was counted 3! = 6 times: ABC, ACB, BAC, BCA, CAB, CBA
Divide: $60 \div 6 =$ **10** committees

Combination formula: ${}_nC_r = \binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$ choose r things from n things

${}_nC_r$ and $\binom{n}{r}$ are both read "n choose r." Some books write $C(n, r)$.

Example: ${}_5C_3 = \dfrac{5!}{3!\,2!} = \dfrac{120}{6 \cdot 2} = 10$

Permutation or combination? Ask: *If I switch the order, is it a different result?*

Order matters: permutation	**Order does not matter: combination**
President, vice president, secretary (different jobs)	A committee of 3 (same job)
1st, 2nd, 3rd place	Picking 3 winners of the same prize
Arranging books on a shelf	Choosing 3 books to take on a trip

Handshakes. 10 people each shake hands once with everyone else. A handshake is a pair, and Ana-Ben is the same as Ben-Ana: ${}_{10}C_2 = \dfrac{10 \cdot 9}{2} = 45$ handshakes.

Work with a partner. Compare answers when you finish each problem.

1. Find each value.

   (a) ${}_5C_2$ $\dfrac{5 \cdot 4}{2 \cdot 1} = 10$

   (b) ${}_6C_3$ $\dfrac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = 20$

   (c) ${}_8C_1$ 8

   (d) $\binom{7}{7}$ 1 (there is one way to choose all of them)

2. Permutation or combination? Then find the number of ways.

   (a) Choose 3 pizza toppings from 10. Combination. ${}_{10}C_3 = 120$

   (b) Give 1st, 2nd, and 3rd prizes to 3 of 10 people. Permutation. ${}_{10}P_3 = 720$

   (c) Choose a president and a treasurer from 12 people. Permutation. ${}_{12}P_2 = 132$

   (d) Choose 4 people from 15 for a committee. Combination. ${}_{15}C_4 = 1{,}365$

3. 12 people meet. Each person shakes hands once with every other person. How many handshakes?

   ${}_{12}C_2 = \dfrac{12 \cdot 11}{2} = 66$

4. A restaurant has 8 toppings. A pizza has 2 different toppings. How many different 2-topping pizzas are there?

   ${}_8C_2 = \frac{8 \cdot 7}{2} = 28$

5. Compare ${}_{10}P_3 = 720$ and ${}_{10}C_3 = 120$.

   (a) Why is ${}_{10}P_3$ bigger? It counts each group of 3 once for every order.

   (b) What number do you divide by to get from 720 to 120? Why? 3! = 6. Each group of 3 can be ordered 6 ways.

6. A hand of 5 cards is dealt from a deck of 52 cards. Use a calculator. How many different hands are there?

   ${}_{52}C_5 = 2{,}598{,}960$

7. A student says, “Choosing 2 of 6 students to clean the board is a permutation, because there are 2 people.” Do you agree? Explain.

   No. Both students do the same job, so the order does not matter. It is a combination: ${}_6C_2 = 15$.